

Improving Mathematical Reasoning in LLMs via Formal Specification Injection: A Study on GSM-Symbolic

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Abstract

Mathematical problem solving has long served as a benchmark for evaluating the reasoning capacity of artificial intelligence systems. With the rise of large language models (LLMs), this question has taken on renewed urgency: do these models genuinely reason, or do they rely on pattern matching over training data? This work replicates and extends the GSM-Symbolic benchmark, which probes mathematical reasoning by systematically varying the numerical values and name variables of mathematical word problems. Three prompting strategies are used to evaluate the performance of the model: standard 8-shot Chain-of-Thought, formal specification with template examples, and formal specification without template examples. The results confirm the previous findings that model performance degrades consistently as question difficulty increases, and that this degradation persists even when the model is explicitly instructed to derive a formal mathematical specification of the problem before solving it. Although formal specification injection offers a modest improvement on moderately harder variants, it provides little benefit on the hardest setting. These findings suggest that prompting alone is not sufficient to address the underlying limitations of current LLMs in mathematical reasoning.

<https://github.com/xiaomanong007/gsm-symbolic-eval>

1 Introduction

More than half a century ago, Alan Turing, in his famous paper *Computing Machinery and Intelligence* (?), posed a question that endures to this day: “Can machines think?” To investigate into this question, Turing proposed the Imitation Game, in which a human interrogator is required to communicate with an agent through text only, without direct access to the agent’s physical form. The interrogator’s task is thus to determine, from the

content of the exchanged information, whether the agent behind the scene is a human or a machine. Remarkably, this setup closely mirrors how modern large language models are deployed in practice: users interact with these systems or machines entirely through a text interface, with no visibility into the underlying computation performed.

With the advancement of large language models, Turing’s question can now be revisited in earnest. These models exhibit impressive performance on creative tasks such as writing and image generation. Yet to substantiate the claim that the machine possesses genuine reasoning capacity, it is necessary to evaluate it on tasks that demand structured, multi-step inference rather than pattern completion. Mathematical problem solving provides a natural benchmark for this purpose, as it requires not only language comprehension to the questions given, but also the ability to identify relevant information, construct a solution plan, and execute it correctly through logical inferences.

This work attempts to shed light on Turing question by using an empirical study of mathematical reasoning in large language models, examining where these models succeed and where they fail. The framework used to assess reasoning capacity is GSM-Symbolic (?), an improved benchmark designed to measure model performance with greater control than its predecessor GSM8K. For GSM-Symbolic, questions are divided into 50 instance sets, each containing 100 questions generated from the same 100 symbolic templates. This setup provides a more reliable estimate of performance of LLM via mean and standard deviation across different sets. Moreover, two harder variants — GSM-P1 and GSM-P2 — are introduced into the experiment. These two sets contain one and two irrelevant clauses, and can be used to test whether models can identify and discard distracting information while maintaining correct reasoning over the relevant content.

To further investigate the reasoning capacity of the model, two additional experiments are conducted in which the model is required to derive a formal mathematical specification of the problem before attempting a solution. Further, the model is also asked to solve the question based on that derived specification rather than the original natural language text. The underlying conjecture is that reasoning in a formal mathematical language may allow the model to focus on the structural relationships within the problem, rather than being misled by surface-level patterns or memorised solution templates encountered during pre-training on GSM8K-style data.

2 Related Work

Mathematical reasoning in LLMs has been extensively studied since the introduction of GSM8K, a standard benchmark for grade-school arithmetic reasoning. ? demonstrated that Chain-of-Thought prompting substantially improves model performance on such tasks, and has since become the standard evaluation strategy. ? subsequently identified a key limitation of fixed benchmarks: models may have encountered test questions during pre-training, and this may inflate reported performance. Their GSM-Symbolic benchmark addresses this through template-based generation, revealing that performance drops significantly when numerical values or problem surface forms are altered — even when the underlying reasoning structure is unchanged.

More broadly, ? survey the current state of mathematical reasoning in LLMs, concluding that despite impressive performance on structured benchmarks, these models may emulate rather than genuinely represent logical reasoning, and that natural-to-formal translation remains error-prone. These findings directly motivate the formal specification experiments conducted in this work.

3 Methodology

3.1 GSM8K and GSM-Symbolic

As mentioned above, GSM8K (?) is a benchmark of 8,500 grade school math word problems, and GSM-Symbolic (?) extends this by generating symbolic variants from templates, randomising names and numerical values while preserving the underlying reasoning structure. Table 1 illustrates the relationship between an original GSM8K question and its GSM-Symbolic counterpart.

The central motivation of this benchmark design is to probe the genuine reasoning capacity of large language models. For a human solver, changing the names or numerical constants in a problem, or both, should not impair the comprehension of the problem, since a competent reasoner grasps the underlying mathematical relationships between quantities rather than the surface form of the text. Consequently, if a model under evaluation shows little or no performance drop under such variation, one may conclude that it has internalised the implicit logical structure of the problem, rather than relying on superficial pattern matching over natural language.

Tables 2 illustrate the progression from GSM-Symbolic to the P1 variants. The answer remains unchanged across all variants; only the distracting context increases. The central idea is to further probe the model’s reasoning capacity: specifically, whether it can correctly identify relevant information when redundant or misleading clauses are introduced. If a model maintains consistent performance across these variants, it provides stronger evidence that the model genuinely comprehends the question and reasons from the relevant information alone, rather than being distracted by irrelevant context.

3.2 Model

All experiments in this work are conducted using GPT-4o-mini. While stronger models exist for mathematical reasoning — such as DeepSeek-Prover — GPT-4o-mini was chosen due to resource constraints. Nevertheless, the writer believe that other Transformer-based language models should have the same trends as that of GPT-4o-mini. Researchers wishing to replicate these experiments with alternative models may do so using the accompanying codebase,¹ in which the model can be configured via a single environment variable.

3.3 Prompt, Chain-of-Thought, and Evaluation

Prior work has demonstrated that Chain-of-Thought (CoT) prompting (?) substantially improves the reasoning performance of large language models by eliciting explicit intermediate steps. To maximise model performance across all experiments, 8-shot examples are incorporated into the CoT prompting (for details on shot construction,

¹<https://github.com/xiaomanong007/gsm-symbolic-eval>

GSM8K (Original)	GSM-Symbolic (Generated)
Benny saw a 10-foot shark with 2 6-inch remoras attached to it. What percentage of the shark’s body length is the combined length of the remoras?	Olivia saw a 210-foot whale with 7 72-inch remoras attached to it. What percentage of the whale’s body length is the combined length of the remoras?
Answer: 10%	Answer: 20%
10 Numerical value changed	Olivia Name changed

Table 1: Comparison of an original GSM8K question and its GSM-Symbolic variant. Names and numerical values are randomised while the underlying reasoning structure remains identical.

GSM-Symbolic (Base)	GSM-P1 (One Extra Clause)
Olivia saw a 210-foot whale with 7 72-inch remoras attached to it. What percentage of the whale’s body length is the combined length of the remoras?	Olivia saw a 210-foot whale with 7 72-inch remoras attached to it. The whale was swimming alongside a 85-foot dolphin. What percentage of the whale’s body length is the combined length of the remoras?
Answer: 20%	Answer: 20% (extra clause is irrelevant)
20 Numerical value	The whale was swimming alongside a 85-foot dolphin. Extra clause added (P1)

Table 2: GSM-Symbolic base variant vs. GSM-P1, which adds one extra clause containing an irrelevant numerical value. The correct answer is unchanged, but the model must identify which information is relevant.

see Section 3.4), instructing the model to reason step by step and produce intermediate results before stating the final answer. While the evaluation metric considers only the correctness of the final numerical answer, the intermediate reasoning chain provides valuable diagnostic information — particularly for identifying where the model is distracted by the irrelevant clauses introduced in GSM-P1 and GSM-P2.

The prompt template used across all experiments is shown below (for a visual comparison of all prompt structures, see Figure 7 in the Appendix):

```

As an expert problem solver, solve step
by step
the following mathematical questions.
Q: {shot question}
A: Let's think step by step.
{reasoning}.
The final answer is {answer}.

:
(8 shot examples)

Q: {target question}
A: Let's think step by step.

```

3.4 Shots and Formal Specification

The shot examples included in each prompt are drawn randomly from either GSM8K or GSM-Symbolic, depending on the experiment. There are two motivations for incorporating shots.

First, as noted in the previous section, CoT prompting improves reasoning performance. Beyond this, providing a fixed answer template across

all shots makes it straightforward to extract the final answer from the model’s response. Specifically, each shot answer terminates with the phrase The final answer is X, and the model learns to replicate this format. The predicted answer is then parsed by searching for this pattern using a regular expression, falling back to the last number appearing in the response if the pattern is absent.

Second, shot prompting draws an analogy to the human learning process. By observing a set of examples — each pairing a question with its step-by-step solution — the agent learns both to interpret the question and to replicate the reasoning pattern. One may also draw a loose analogy to fine-tuning, in which a model is adapted toward preferred (input, output) pairs; however, since shot prompting operates entirely at inference time without updating model weights, the connection is superficial rather than mechanistic.

A *formal specification* is defined as a mathematical translation of a natural language question into an explicit symbolic representation. Concretely, given a question such as:

Mei saw a 90-foot dolphin with 3 36-inch remoras attached to it. What percentage of the dolphin’s body length is the combined length of the remoras?

the corresponding formal specification, extracted directly from the GSM-Symbolic template, takes the form:

```

#init:
- dolphin_length = 90    (feet)
- remora_length = 36     (inches)
- num_remoras = 3

#conditions:
- combined = num_remoras × remora_length
- ans = (combined / (dolphin_length × 12)) × 100

#answer: ans

```

Here, `#init` enumerates the variables and their values as stated in the question, and `#conditions` encodes the mathematical relationships between them (the formula by which the question is actually solved).

This two-stage process — first translating a problem into a formal representation, then solving from that representation — reflects a habit of human reasoners, and is often cited as a distinguishing marker between genuine comprehension and superficial pattern matching. Several studies have shown that prompting models to perform this translation step can improve reasoning performance, though the translation itself is not always correct (?).

4 Experiment

4.1 Baseline

The purpose of replicating the experimental results of ? is twofold: to verify the correctness of the reported findings, and to validate the experimental pipeline used in this work. Table 3 presents the results from the original paper, and the first block of Table 4 shows the replicated results.

Across all three variants, the replicated results are consistently lower than those reported in the paper, with the gap being most pronounced on GSM-P2, where the difference reaches approximately 20%. The writer attribute this discrepancy primarily to the output token size. The baseline is configured with a maximum of 512 output tokens, which may be insufficient for GSM-P2 questions: since extra irrelevant clauses are appended to the target question, the model must process additional context, leading to longer reasoning chains and a higher risk of truncation before a final answer is produced. The subsequent experiments address this by increasing the token budget to 1024. For the purposes of replication, however, the current results are sufficient to confirm the overall trend, and readers may discount the GSM-P2 baseline figures without affecting the general conclusions of this work.

The results in the above two tables provide sufficient evidence to conclude that model performance degrades when irrelevant clauses are added. Mirzadeh attributes this degradation primarily to "data contamination": since portions of GSM8K question may have been seen during pre-training, the model may rely on memorised solutions rather than genuine reasoning, and this reliance breaks down when the problem structure is complexed by additional clauses.

This explanation is plausible, yet the writer argues that it is not sufficient to explain the phenomena. Drawing an analogy to human reasoning, even a competent human solver may perform worse when irrelevant information is introduced, as it demands additional cognitive effort to identify which clauses are useful and which are distracting. The performance drop may therefore reflect a general limitation of sequential reasoning under noise, for any agent whatsoever. A more complete account would need to disentangle the contribution of memorisation from the inherent difficulty of filtering irrelevant context during multi-step reasoning.

4.2 Formal Specification

The observation above motivates the following two experiments, in which the model is first asked to derive a formal mathematical specification of the problem before solving it. The hypothesis being that an explicit formalisation step "forces" the model to actively identify the relevant variables and relationships, rather than pattern-matching on the surface form of the question.

The two experiments differ primarily in their prompt construction. First, the system header is augmented with the following instruction:

For each question you will see a formal specification that defines the variables and mathematical relationships in the problem. Study how the formal specification maps to the question and how it is used in the solution.

For the final question, NO formal specification is provided. You must:

1. Derive the formal specification yourself (`#init` and `#conditions`)
2. Use your derived specification to solve the problem step by step
3. State the final answer

Compared to the baseline, steps 1 and 2 explicitly instruct the model to first translate the natural language question into a formal mathematical specification, and then solve the problem from that phenomena rather than from the original text directly.

Model	GSM8K (FULL)	GSM-Symbolic	GSM-P1	GSM-P2
GPT-4o	95.2	94.9 $\pm$ 1.87	93.9 $\pm$ 2.59	88.0 $\pm$ 3.43
GPT-4o-mini	94.2	91.7 $\pm$ 2.02	81.1 $\pm$ 3.05	72.4 $\pm$ 4.57

Table 3: Results from ? Tab. 1. All values are accuracy (%). Mean $\pm$ std computed across 50 instance sets.

Method	Variant	Mean	Std	Min	Max	N
Baseline	GSM-Symbolic	86.3	2.3	81.0	90.0	50
	GSM-P1	78.5	2.5	73.0	83.0	50
	GSM-P2	51.1	4.7	40.0	64.0	50
Formal (w/ template)	GSM-Symbolic	83.2	1.8	80.0	87.0	25
	GSM-P1	77.9	2.8	70.0	83.0	25
	GSM-P2	62.8	5.0	54.0	72.0	25
Formal (no template)	GSM-Symbolic	85.9	2.2	81.0	89.0	25
	GSM-P1	81.2	2.6	76.0	86.0	25
	GSM-P2	70.2	3.1	62.0	76.0	25

Table 4: Our results using gpt-4o-mini, 8-shot CoT, greedy decoding. Mean $\pm$ std computed across N instance sets. All values are accuracy (%).

Second, the shot examples differ between the two experiments. In the *formal with template* experiment, each shot is accompanied by its formal specification extracted from the GSM-Symbolic template (since GSM-Symbolic provides ground-truth formal specifications for all questions, the shots are drawn from the GSM-Symbolic test set rather than GSM8K). In the *formal without template* experiment, the shots are identical to those used in the baseline (standard GSM8K examples with no formal specification attached). This means that the model must infer the expected formalisation format itself, without concrete examples to guide it.

The second and third blocks of Table 4 present the results for the two formal specification experiments. Both experiments show a consistent performance decline as the difficulty of the variant increases, suggesting that additional irrelevant clauses continue to impair model performance regardless of the prompting strategy.

However, it is worth noting that the performance drop from GSM-Symbolic to GSM-P1 is smaller in both formal experiments than in either the baseline or the results reported by Mirzadeh. Specifically, Mirzadeh reported a drop of approximately 10% from GSM-Symbolic to GSM-P1, whereas both formal specification experiments exhibit a drop of approximately 5%.

Two explanations may account for this improvement. First, consistent with the hypothesis motivating these experiments, the explicit formalisation

step may encourage the model to identify relevant variables and relationships more carefully, thereby making it more robust to irrelevant clauses. Or second, the increased output token budget (1024 tokens, compared to 512 in the baseline) affords the model more reasoning steps, which may independently contribute to improved performance on harder variants.

Both explanations are plausible. But ? note that natural-to-formal translation is itself error-prone, with models liable to misdefine variables or introduce redundant definitions. These errors during the translation will corrupt the subsequent reasoning chain regardless of its length.

The results for GSM-P2 show little improvement across both formal specification experiments relative to the baseline. This suggests that, when two irrelevant clauses are present, the formalisation step alone is not sufficient to fully resolve the added difficulty.

Figure 1 provides a direct visual comparison of all three experiments across the three variants, illustrating both the consistent downward trend in performance as variant difficulty increases and the relative positioning of each prompting strategy.

5 Conclusion

This work replicates and extends the findings of Mirzadeh’s, confirming that current large language models exhibit persistent limitations in mathematical reasoning that are not easily remedied by prompt engineering alone. Even when an explicit

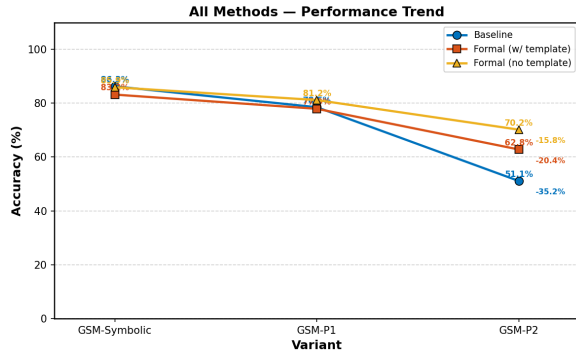


Figure 1: Accuracy of all three prompting methods across the three GSM-Symbolic variants. Each point represents the mean accuracy across instance sets, with dashed lines indicating the performance trend. The downward slope for all methods confirms that added irrelevant clauses consistently impair model performance.

formal translation step is introduced as an intermediate guide toward the final solution, performance continues to degrade as question difficulty increases. In some cases, merely altering the numerical values within the same question template is sufficient to impair the model’s ability to produce a correct answer.

6 Further Thoughts

Although not strictly relevant to the central content of this paper, the writer has found some interesting ideas in Turing’s paper that merit brief discussion. Turing proposed three components relevant to imitating an adult human mind (?):

- (a) The initial state of the mind, say at birth.
- (b) The education to which it has been subjected.
- (c) Other experience, not to be described as education, to which it has been subject.

He then proposed that, “instead of trying to produce a programme to simulate the adult mind, why not rather try to produce one which simulates the child’s?”²

It is worth considering which state the current LLM falls into: the “adult’s mind” or the “child’s mind”. The conventional training pipeline described by ? consists of:

1. Pre-training;

²For Turing, constructing instruction tables is usually described as “programming.” To “programme a machine to carry out the operation A” means to put the appropriate instruction table into the machine so that it will do A (?).

2. Supervised Fine-Tuning (SFT);
3. Reward Model Training;
4. Reinforcement Learning (RL);
5. Optional additional SFT to mitigate RL side-effects.

Comparing the two frameworks, it is apparent that stages (2)–(5) correspond to components (b) and (c) proposed by Turing. The question then becomes whether stage (1) and component (a) are equivalent. The writer can not offer a definitive answer, but he believes this points toward a fruitful direction for further study: how the “initial state” of a machine should be determined, to what extent the “mechanical mind” should imitate the human mind, and in which respects it should not.

A Additional Plots

A.1 Statistical Plots

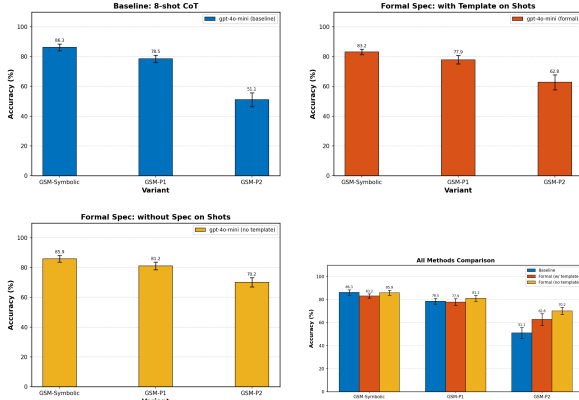


Figure 2: Accuracy per method across variants. Top left: Baseline. Top right: Formal with template. Bottom left: Formal without template. Bottom right: all methods overlaid.

These plots show the mean accuracy $\pm 1\sigma$ for each prompting method across the three GSM-Symbolic variants. The consistent leftward shift across variants confirms the performance degradation as irrelevant clauses are added. Both formal specification experiments exhibit a smaller drop from GSM-Symbolic to GSM-P1 than the baseline, though all three methods converge on GSM-P2.

A.2 Distribution Plots

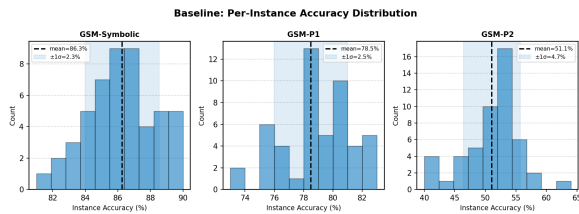


Figure 3: Baseline per-instance accuracy distribution.

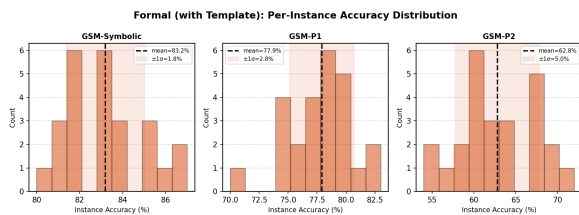


Figure 4: Formal with template per-instance accuracy distribution.

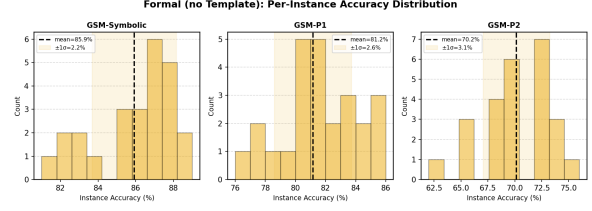


Figure 5: Formal without template per-instance accuracy distribution.

These histograms show the spread of per-instance accuracy values across instance sets for each method and variant. A narrow, rightward distribution indicates consistently high performance; a wide or leftward distribution reflects high variance or low accuracy. The baseline distributions widen significantly on GSM-P2, while both formal experiments show tighter distributions on GSM-P1.

A.3 Question Type Analysis

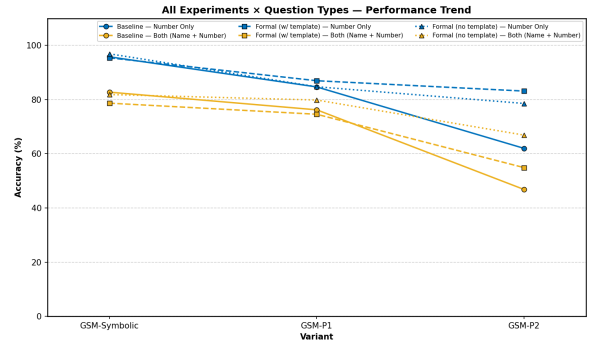


Figure 6: Performance trend by question type across all experiments.

Questions in GSM-Symbolic fall into two categories: those where only numerical values were changed (*number only*), and those where both names and numbers were changed (*both*). This plot compares performance across these two types for all three experiments. The near-identical trends suggest that name substitution has little effect on model performance.

A.4 Prompt Structure

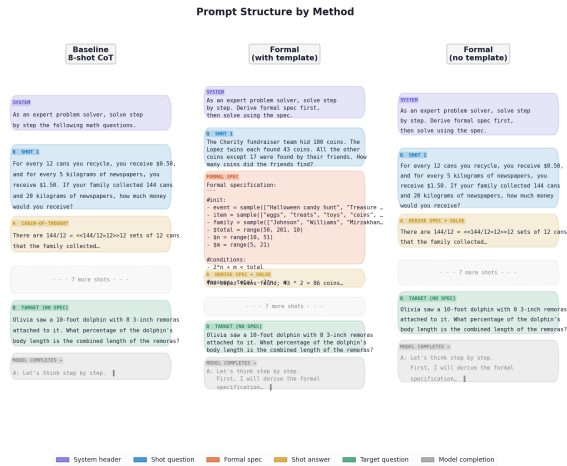


Figure 7: Prompt structure comparison across all three methods.

Figure 7 illustrates the full prompt structure for each method side by side. The baseline uses a compact 8-shot CoT format. The formal with template experiment extends each shot with a symbolic specification block, resulting in a substantially longer prompt. The formal without template experiment shares the same structure as the baseline in the shot portion but modifies the system header and the target question stub.